# Finite terms $(\Lambda \to 0)$

updated: reality, Row III logarithms at finite $\vee$, Group II and III simplified

## 1 Group I (with c.c.)

**Reality.** Each term here is $T = \int e^{-ik\cdot\Delta}\, c\,\mathcal{K}\,\mathcal{O}$ with a real kernel $\mathcal{K}$, and the c.c. is added. For classical $\rho$: if $c$ is real, $T + T^\dagger = 2c \int \cos(k\cdot\Delta)\,\mathcal{K}\,\mathcal{O}$; if $c = i\alpha$ with $\alpha$ real, $T + T^\dagger = 2\alpha \int \sin(k\cdot\Delta)\,\mathcal{K}\,\mathcal{O}$, since $i\,e^{-ik\cdot\Delta} + \text{c.c.} = 2\sin(k\cdot\Delta)$. So every explicit $i$ disappears. Throughout, $\kappa \equiv k^+/\vee \in (0,1)$ and $u \equiv (x-z)^2/(x-w)^2$. The arguments $(1-\kappa)u+\kappa$ and $1-\kappa$ of the logarithms in Rows II, IV and V are positive. The only logarithms with possibly negative arguments are in Row III; they are treated there.

### 1.1 Row I

NA

### 1.2 Row II

We write it in two parts:

$$\frac{d^3N_{1\,\text{NLO}}}{d^3k} = +\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left\{-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right\}\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^{b}(x) - U^{ab}(x)\,\rho^{b}(x)\Big] \tag{1.1}$$

$$\frac{d^3N_{2b\,\text{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2q\int_{x,y} e^{iq\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{q}^2}\,R^i(\mathbf{q},\mathbf{p})$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big] \tag{1.2}$$

where

$$R^i = -2q^i\log\left(\frac{p^2}{q^2+\frac{k^+}{\vee}\,p^2}\right)+\frac{(2p^2-2\,q\cdot p+q^2)\,q^i+q^2p^i}{(q-p)^2}\log\left[\frac{q^2}{(q-p)^2}+\frac{k^+}{\vee}\left(1-\frac{q^2}{(q-p)^2}\right)\right]-\frac{p^2q^i+q^2p^i}{(q-p)^2}\log\left(1-\frac{k^+}{\vee}\right). \tag{1.3}$$

The integration momentum is renamed $k \to q$, because it clashed with the external $k$. All logarithms have positive arguments, since $\frac{q^2}{(q-p)^2}+\kappa\big(1-\frac{q^2}{(q-p)^2}\big) = (1-\kappa)\frac{q^2}{(q-p)^2}+\kappa > 0$. Every term of $R^i$ is odd under $(q,p)\to(-q,-p)$, so the double Fourier integral is $i\times$(real). With the c.c. this row is therefore real (sin).

### 1.3 Row III

We write it in three parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2}{(x-w)^2}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{1.4}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.5}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{1.6}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right]\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(w)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{1.7}
$$

where:

$$\mathcal{A}_0 = \frac{1}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \left\{ \arctan \left[ \frac{(x-z)^2 - (x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] \right.$$
$$\left. + \arctan \left[ \frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] \right\} = \frac{\theta_w}{\big|(x-w)\times(z-w)\big|}, \qquad \theta_w = \arccos \frac{(x-w)\cdot(z-w)}{|x-w|\,|z-w|} \in [0,\pi] \tag{1.8}$$

Here $\theta_w$ is the angle of the triangle $(x,z,w)$ at $w$. This uses $(x-z)^2-(x-w)\cdot(x-z) = -(x-z)\cdot(z-w)$ and $|(x-w)\times(x-z)| = |(z-w)\times(x-z)|$.

**Correction to the last term of $N_{3b}$.** At finite $\Lambda$ this term carries $\log \frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2}$, which tends to $2\log\frac{k^+-\Lambda}{\Lambda}$ as $z\to w$. With the kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ it is therefore *non-integrable* at $z\to w$, even at finite $\Lambda$. Its $1/(z-w)^2$ singularity was cancelled *exactly*, for every $\Lambda$, by the original $-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\frac{k^+-\Lambda}{\Lambda}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ term of $N_{3a}$. That term was replaced, by the JIMWLK-consistency requirement verified at operator level, with the kernel $\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2}$ and $U^{bc}(z)\to U^{bc}(w)$. Both terms come from the same $\delta$-piece of the $p^+$ integrand, so the same replacement is applied here (in red above). The corrected term is integrable at $z\to w$: it behaves as $\log(1/|z-w|)/|z-w|$, and its angle average vanishes. This was checked numerically. Without the correction, $r^2\times$(angle average) tends to a nonzero constant, $\propto \log(k^+/\Lambda)$, as $r=|z-w|\to 0$.

### Row III simplified at finite $\vee$: the negative-argument logarithms

Let $u=\frac{(x-z)^2}{(x-w)^2}$ and $\kappa=\frac{k^+}{\vee}$. As in the limiting case, the logarithms come from the energy denominator of $\bar{\mathbb{B}}_3^\dagger$ and are on the *principal branch*, $\log(-A)=\log A+i\pi$ for $A>0$. The four arguments are

$$\begin{aligned}
\log\frac{(x-z)^2-\kappa(x-w)^2}{(x-z)^2-(x-w)^2} &= \log\frac{u-\kappa}{u-1} = \log\frac{|u-\kappa|}{|1-u|} + i\pi\,\theta(u-\kappa)\theta(1-u),\\
\log\frac{\kappa(x-w)^2-(x-z)^2}{(x-w)^2-(x-z)^2} &= \log\frac{\kappa-u}{1-u} = \log\frac{|u-\kappa|}{|1-u|} + i\pi\,\theta(u-\kappa)\theta(1-u),\\
\log\frac{(x-w)^2-(x-z)^2}{(x-w)^2} &= \log(1-u) = \log|1-u| + i\pi\,\theta(u-1),\\
\log\frac{\kappa(x-w)^2-(x-z)^2}{(x-w)^2} &= \log(\kappa-u) = \log|u-\kappa| + i\pi\,\theta(u-\kappa).
\end{aligned} \tag{1.9}$$

The pairs that appear together add up to the same function, for all $u>0$ (check the three regions $u<\kappa$, $\kappa<u<1$, $u>1$):

$$\log\frac{u-\kappa}{u-1} + \log(1-u) = \log\frac{\kappa-u}{1-u} + \log(1-u) = \log(\kappa-u) = \log|u-\kappa| + i\pi\,\theta(u-\kappa). \tag{1.10}$$

- In $N_1$, the second and third terms have the same kernel and give the first pair. The fourth term is already $\log(\kappa-u)$.
- The six terms of $N_2$ with $\log\frac{\kappa-u}{1-u}$ and the first six terms of $N_{3a}$ with $\log(1-u)$ contain the same six tensors with the same coefficients, so they give the second pair.
- The $\log(1-\kappa)$ terms (first term of $N_1$, first two terms of $N_2$) are all $\propto\delta^{ij}$. With $X=x-w$, $Z=x-z$, $W=z-w$, their coefficient is

$$-\log(1-\kappa)\,\frac{(X^2+Z^2)W^2+2(W\cdot Z)X^2-2(W\cdot X)Z^2}{W^2(X^2-Z^2)^2} = -\log(1-\kappa)\,\frac{(X^2+Z^2)^2-4X^2Z^2}{W^2(X^2-Z^2)^2} = -\frac{\log(1-\kappa)}{W^2}.$$

With the same tensor identity as in the limiting case, all logarithmic terms of $N_1+N_2+N_{3a}$ collapse to

$$2\log(\kappa-u)\,\mathcal{K}^{ij}_{\rm IV} - \log(1-\kappa)\frac{\delta^{ij}}{(z-w)^2} = 2\log\frac{|u-\kappa|}{1-\kappa}\,\mathcal{K}^{ij}_{\rm IV} + 2\log(1-\kappa)\,\widehat{\mathcal{K}}^{ij}_{\rm IV} + 2i\pi\,\theta(u-\kappa)\,\mathcal{K}^{ij}_{\rm IV}, \tag{1.11}$$

contracted with $\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}$. This was checked numerically from the terms above, in all three regions of $u$ and for several $\kappa$, to $10^{-14}$. Here

$$\mathcal{K}^{ij}_{\rm IV} = \frac{\delta^{ij}}{2(z-w)^2} + \widehat{\mathcal{K}}^{ij}_{\rm IV}, \qquad \widehat{\mathcal{K}}^{ij}_{\rm IV} = \frac{(z-w)^i(x-z)^j}{(z-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} - \frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2}. \tag{1.12}$$

All the $\frac{1}{(x-z)^2-(x-w)^2}$ poles cancel. Hence, with $X_\perp = X - \frac{X\cdot Z}{Z^2}Z$ and $Z_\perp = Z - \frac{X\cdot Z}{X^2}X$, Row III $= N_1 + N_2 + N_{3a} + N_{3b}$ becomes

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}\Big|_{\rm III} = \frac{1}{(2\pi)^3}\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\Bigg\{\frac{1}{(x'-w')^2(y-z)^2}\Bigg[ \\
\Big(2\log\frac{|(x-z)^2-\kappa(x-w)^2|}{(1-\kappa)(x-w)^2}\,\mathcal{K}^{ij}_{\rm IV} + 2\log(1-\kappa)\,\widehat{\mathcal{K}}^{ij}_{\rm IV}\Big)(x'-w')^i(y-z)^j \\
+\frac{2\,\theta_w}{|(x-w)\times(z-w)|\,(z-w)^2}\Big((x'-w')\cdot(z-w)\;(y-z)\cdot X_\perp + (x'-w')\cdot Z_\perp\;(y-z)\cdot(z-w)\Big) \\
+\frac{1}{(z-w)^2}\log\frac{(z-w)^2}{(x-w)^2}\Big(\frac{(x'-w')\cdot(x-w)\;(y-z)\cdot(z-w)}{(x-w)^2} - \frac{(x'-w')\cdot(z-w)\;(y-z)\cdot(x-z)}{(x-z)^2}\Big)\Bigg] \\
\times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\Big] \\
-\log\frac{(z-w)^2}{(x-w)^2}\,\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2} \\
\times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{de}(x)U^{bc}(w)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\Big]\Bigg\} + \frac{d^3N_{i\pi}}{d^3k}. \qquad (1.13)
\end{aligned}$$

The $2i\pi\,\theta(u-\kappa)$, multiplied by the overall $i$, gives a *real* term:

$$\begin{aligned}
\frac{d^3N_{i\pi}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4 f^{abd}}{8\pi^4}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\theta\big((x-z)^2-\kappa\,(x-w)^2\big)\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\,\mathcal{K}^{ij}_{\rm IV} \\
\times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\Big]\,. \qquad (1.14)
\end{aligned}$$

**Reality and finiteness.**

- The first expression has coefficient $i\times$real; with its c.c. it becomes $\frac{g^4 f^{abd}}{8\pi^5 k^+}\int \sin(k\cdot(w'-w))\{\cdots\}$.
- $N_{i\pi}$ has a real coefficient; with its c.c. it becomes $-\frac{g^4 f^{abd}}{4\pi^4 k^+}\int\cos(k\cdot(w'-w))\{\cdots\}$.
- No logarithm of a negative number is left.
- The first expression is manifestly integrable at $z\to w$: there $u\to 1$, so $\log\frac{|u-\kappa|}{1-\kappa} = O(|z-w|)$ kills the $\frac{\delta^{ij}}{2(z-w)^2}$ of $\mathcal{K}_{\rm IV}$, and $\widehat{\mathcal{K}}_{\rm IV} = O(1/|z-w|)$. Before the combination, the $\frac{\delta^{ij}}{(z-w)^2}$ pieces of $N_1$, $N_2$ ($\log(1-\kappa)$) and of the logarithms ($\log(1-\kappa)$ at $u=1$) are separately non-integrable; only their sum is finite.
- At $x\to z$ ($u\to 0$) the logarithm tends to $\log\frac{\kappa}{1-\kappa}$: finite at finite $\vee$. The kernel singularity there is only $1/|x-z|$.
- As $\kappa\to 0$ this reduces to the limiting-case result ($\log u + i\pi$, $\theta\to 1$).
- The only singular piece of $N_{i\pi}$ is $\frac{\delta^{ij}}{2(z-w)^2}$ at $z\to w$, where $\theta = 1$. As in the limiting case, the colour factor there is antisymmetric under $w\leftrightarrow w'$ and multiplies the symmetric $\cos(k\cdot(w'-w))$. It cancels against the c.c.: it is a Coulomb phase.

## 1.4 Row IV

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ \\
\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
\left. -\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\
+\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big] \qquad (1.15)
\end{aligned}$$

Positive logarithm ($(1-\kappa)u+\kappa > 0$); with the c.c. this row is $\frac{1}{(2\pi)^3}\frac{g^4}{4\pi^5}\frac{1}{k^+}\int\sin(k\cdot(w'-w))\{\cdots\}$. Compact form: with

$$V^j \equiv \frac{(z-w)^j}{(z-w)^2} + \frac{(x-z)^j}{2\,(x-z)^2}, \qquad \left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} + \frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right] = \frac{(x-w)^i\,V^j}{(x-w)^2}, \qquad (1.16)$$

the kernel is $\log[(1-\kappa)u+\kappa]\,\mathcal{K}^{ij}_{\rm IV} + \log(1-\kappa)\,\frac{(x-w)^iV^j}{(x-w)^2}$, where $\mathcal{K}_{\rm IV}$ is the Row III kernel. At $z\to w$, $\log[(1-\kappa)u+\kappa] = O(|z-w|)$, so every piece is at most $1/|z-w|$: integrable. The same kernel appears in Group II Row I (first part) and Group III Rows I, II.

## 1.5 Row V

**First part**

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\\
&\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+\\
&\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
\end{aligned}
\tag{1.17}
$$

**Second part**

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.18}
$$

$$\frac{d^3N_{2b\,\mathrm{rest}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}\Bigg\{e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}-\left[\frac{k^+}{k^+-p^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right]$$
$$-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left.-e^{-ik\cdot(w'-x)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}-2\right]\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2}\right\}$$
$$\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.19}$$

# 2 Group II

**Reality.** There is no c.c.; each row is hermitian by itself. All logarithms have positive arguments, $(1-\kappa)u+\kappa>0$ and $1-\kappa>0$ with $\kappa=k^+/\vee$, so nothing imaginary is generated.

- Row I, first part: $N_{1b}=N_{1a}^\dagger$ (take the conjugate and relabel $w\leftrightarrow w'$). So $N_{1a}+N_{1b}=-\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int\sin(k\cdot(w'-w))\{\cdots\}$, and its kernel is the Group I Row IV kernel.
- Row I, second part ($N_{2a}$–$N_{2f}$): the coefficients are real. The combined kernel below is symmetric under $(x,w)\leftrightarrow(x',w')$, and so is the colour factor; this gives $\cos(k\cdot(w'-w))$.
- Row IV is symmetric under $(x,y)\leftrightarrow(x',y')$, which gives $\cos(k\cdot(y'-y)/\xi)$.

## 2.1 Row I

### 2.1.1 First part

$$\frac{d^3N_{1a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{16\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$\left.-\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\right\}$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{2.1}$$

$$
\begin{aligned}
\frac{d^3 N_{1b\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{x,y,z,w,x',w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{(x-w')^2}\right]\times \\
& \left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right] \\
& +\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\} \\
& \times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right.
\end{aligned}
\tag{2.2}
$$

Compact form: the kernels of $N_{1a}$ and $N_{1b}$ are $\log[(1-\kappa)u+\kappa]\,\mathcal{K}^{ij}_{\mathrm{IV}}+\log(1-\kappa)\frac{(x-w)^i V^j}{(x-w)^2}$ (the Group I Row IV kernel), and the same with $w\to w'$. They are integrable at $z\to w$ (resp. $z\to w'$) and at $x\to z$.

### 2.1.2 Second part

$$
\begin{aligned}
\frac{d^3 N_{2a\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\Bigg\{ & \\
& \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times \\
& \left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right. \\
& +\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2} \\
& +\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2} \\
& \left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times \\
& \left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\Bigg\}\times \\
& \left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
& \left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{2.3}
$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$

$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$

$$+\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\;-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.4}$$

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{2.5}
$$

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} &= +\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{\left[(x-z)^2-(x-w)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{\left[(x'-w')^2-(x'-z)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{2.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{2.7}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\big[(x-z)^2-(x-w)^2\big]}\\
&\quad\frac{\big[(x'-z)^2-(x'-w')^2\big]}{\big[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\big]}\times\log\left[\frac{\Big(1-\frac{k^+}{\vee}\Big)\,(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\quad\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\left[\frac{1}{\vee}-\frac{1}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}\\
&\quad\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\quad\frac{\big[(x-z)^2-(x-w)^2\big]}{\big[(x'-z)^2-(x'-w')^2\big]\,\big[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\big]}\times\log\left[\frac{\Big(1-\frac{k^+}{\vee}\Big)\,(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\quad\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{2.8}
$$

**Second part simplified: $N_{2a}+\cdots+N_{2f}$ in one kernel**

All twelve pieces of $N_{2a}$–$N_{2f}$ carry the same colour factor

$$
\begin{aligned}
\mathcal{C}_2 &= f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')U^{cb}(z)U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\,\rho^{e'}(x')\rho^{d}(x)+N_c\,U^{de'}(x')U^{de}(x)\,\rho^{e'}(x')\rho^{e}(x).
\end{aligned}
\tag{2.9}
$$

This colour factor does *not* vanish at $z=w$ or at $z=w'$. We checked this numerically with random SU(3) adjoint matrices. So every $\frac{1}{(z-w)^2}$ or $\frac{1}{(z-w')^2}$ in the kernel has to cancel between the pieces. In particular, the middle term of $N_{2f}$, $\frac{d_\perp}{4}\,(\kappa-1)\,\frac{1}{(z-w)^2(z-w')^2}$, is non-integrable at both points on its own.

**Notation.** $X=x-w$, $Z=x-z$, $W=z-w$, $X'=x'-w'$, $Z'=x'-z$, $W'=z-w'$, so that $X=Z+W$ and $X'=Z'+W'$. Further,

$$
\begin{aligned}
&u=\frac{Z^2}{X^2},\quad u'=\frac{Z'^2}{X'^2},\quad f(s)=\log\big[(1-\kappa)s+\kappa\big],\quad \Delta\big[F(s)\big]\equiv\frac{F(u)-F(u')}{u-u'},\\
&V=\frac{W}{W^2}+\frac{Z}{2Z^2},\quad P=\frac{W}{W^2}-\frac{X}{2X^2},\quad V'=\frac{W'}{W'^2}+\frac{Z'}{2Z'^2},\quad P'=\frac{W'}{W'^2}-\frac{X'}{2X'^2}.
\end{aligned}
\tag{2.10}
$$

The bracket of $N_{2a}$, the bracket of $N_{2b}$, the bracket of $N_{2c}$, the bracket of $N_{2d}$ and the bracket of $N_{2e}$ factorise as

$$
\begin{aligned}
&\mathcal{A}=(X\cdot P')(Z'\cdot V)+(X'\cdot P)(Z\cdot V'),\qquad \mathcal{B}=(X\cdot X')\,(V\cdot V'),\qquad \mathcal{C}=(Z\cdot Z')\,(P\cdot P'),\\
&\mathcal{D}=\frac{(Z'^2-X'^2)\,X\cdot V}{2W'^2}+\frac{(Z^2-X^2)\,X'\cdot V'}{2W^2},\qquad \mathcal{E}=\frac{(Z'^2-X'^2)\,Z\cdot P}{2W'^2}+\frac{(Z^2-X^2)\,Z'\cdot P'}{2W^2}.
\end{aligned}
\tag{2.11}
$$

With $(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2=X^2X'^2(u-u')$, each piece becomes a divided difference:

- $N_{2a}\to\mathcal{A}\,\Delta[f]$;
- $N_{2b}$: the first term $\to\log(1-\kappa)\,\mathcal{B}$ exactly, because $\big[(x-z)^2-\frac{(x-w)^2(x'-z)^2}{(x'-w')^2}\big]$ cancels the denominator; the second and third terms $\to-\mathcal{B}\,\Delta[sf]$;
- $N_{2c}\to-\mathcal{C}\,\Delta[f/s]$;
- $N_{2d}\to\mathcal{D}\,\Delta[\frac{sf}{s-1}]$;
- $N_{2e}\to-\mathcal{E}\,\Delta[\frac{f}{s-1}]$;

- $N_{2f}$, all three terms together:

$$-\frac{d_\perp}{4}\frac{1}{W^2W'^2}\Big[(u-1)(u'-1)\,\Delta\Big[\frac{sf}{(s-1)^2}\Big]+1-\kappa\Big]=-\frac{d_\perp}{4}\frac{(u-1)(u'-1)}{W^2W'^2}\,\Delta\big[\widetilde{F}(s)\big],\qquad \widetilde{F}(s)=\frac{s\,f(s)-(1-\kappa)(s-1)}{(s-1)^2}.$$

  The equality holds because $F(s)=\frac{sf(s)}{s-1}$ is regular at $s=1$ with $F(1)=f'(1)=1-\kappa$, and $(u'-1)F(u)-(u-1)F(u')=(1-\kappa)(u'-u)+(u-1)(u'-1)\big[\widetilde{F}(u)-\widetilde{F}(u')\big]$. This is where the middle term of $N_{2f}$ is absorbed.

Altogether, using $X^2(u-1)=Z^2-X^2$:

$$\begin{aligned}\sum_{n=a}^{f}\frac{d^3N_{2n\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik\cdot(w'-w)}\Bigg\{\\ \frac{1}{(x-w)^2(x'-w')^2}\Big[\mathcal{A}\,\Delta[f]-\mathcal{B}\,\Delta[sf]-\mathcal{C}\,\Delta\Big[\frac{f}{s}\Big]+\mathcal{D}\,\Delta\Big[\frac{sf}{s-1}\Big]-\mathcal{E}\,\Delta\Big[\frac{f}{s-1}\Big]+\log(1-\kappa)\,\mathcal{B}\Big]\\ -\frac{d_\perp}{4}\,\frac{\big[(x-z)^2-(x-w)^2\big]\big[(x'-z)^2-(x'-w')^2\big]}{(x-w)^2(x'-w')^2(z-w)^2(z-w')^2}\,\Delta\big[\widetilde{F}\big]\Bigg\}\,\mathcal{C}_2,\qquad d_\perp=2.\end{aligned}\tag{2.12}$$

This was checked numerically against the sum of the twelve original pieces, at random points and for random $\kappa$, to $10^{-15}$.

**Finiteness and reality.**

- Every function inside $\Delta[\cdot]$ ($f$, $sf$, $f/s$, $\frac{sf}{s-1}$, $\frac{f}{s-1}$, $\widetilde{F}$) is regular at $s=1$, because $f(1)=0$. So $\Delta$ is regular at $u=u'$, where it equals $F'(u)$. All the poles $\frac{1}{(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2}$, $\frac{1}{(x-z)^2-(x-w)^2}$ and $\frac{1}{(x'-z)^2-(x'-w')^2}$ of the original pieces are removable.
- $z\to w$: $(x-z)^2-(x-w)^2=-W\cdot(X+Z)=O(|W|)$, so the last term is $O(1/|W|)$, and so are $\mathcal{A},\dots,\mathcal{E}$: integrable. The same holds at $z\to w'$. Numerically, $r^2\times$(angle average over $z$ on a circle of radius $r$ around $w$ or $w'$) $\propto r^2$, i.e. the angle average is finite. The middle term of $N_{2f}$ alone would give $r^2\langle\cdot\rangle\to\frac{d_\perp}{4}(\kappa-1)/(w-w')^2\neq 0$.
- $x\to z$: $f(0)=\log\kappa$ is finite, $\mathcal{C}\propto Z$, $V=O(1/|Z|)$: at most $1/|Z|$, integrable. The same holds for $x'\to z$.
- $\Delta$ is symmetric under $u\leftrightarrow u'$, and $\mathcal{A},\dots,\mathcal{E}$ and $\mathcal{C}_2$ are symmetric under $(x,w)\leftrightarrow(x',w')$. So the integral equals its complex conjugate: it is real, with $e^{-ik\cdot(w'-w)}\to\cos(k\cdot(w'-w))$.
- As $\kappa\to 0$: $f=\log s$, $\log(1-\kappa)\mathcal{B}\to 0$, $\widetilde{F}=\frac{s\log s-s+1}{(s-1)^2}$. This is the limiting case.

### 2.2 Row II

NA

### 2.3 Row III

NA

### 2.4 Row IV

$$\begin{aligned}\frac{d^3N_{\mathrm{Finite}}}{d^3k}=+\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\;(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\ \times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned}\tag{2.13}$$

## 3 Group III (with c.c.)

**Reality.** All logarithms have positive arguments, and the denominators $p^+(x-z)^2+k^+(x-w)^2$ are positive, so nothing imaginary is generated.

- Rows I, II, III (first part), V and VI have coefficient $i\times$real and give sin with the c.c.
- Row III (second part) is real and gives $2\,\mathrm{Re}$, i.e. the cosine of the total phase.
- Rows I and II have the Group I Row IV kernel, $\log[(1-\kappa)u+\kappa]\,\mathcal{K}^{ij}_{\mathrm{IV}}+\log(1-\kappa)\frac{(x-w)^iV^j}{(x-w)^2}$, which is integrable at $z\to w$ and at $x\to z$.

### 3.1 Row I

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right.$$
$$\left.-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}\right]$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{d'b}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.1}$$

### 3.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right.$$
$$\left.-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}-\frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2}\right]$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{c'b}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.2}$$

### 3.3 Row III

#### 3.3.1 First part

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m(x'-y')^{k'}(y-w)^i(x-z)^j}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+(p^++k^+)}\right]_++\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m(y-w)^i(x-z)^j}{(y'-z)^2(y-w)^2(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\left[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}$$
$$\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{3.3}$$

### 3.3.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\\
&-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_++\frac{1}{(p^++k^+)}\right)\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]\\
&\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times\\
&\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(\right.\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}\\
&\left[\frac{(x-w)^2}{(x-z)^2\,(p^++k^+)\,[p^+\,(x-z)^2+k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.4}
$$

**Simplifications in the second part.**

- With $V^j=\frac{(z-w)^j}{(z-w)^2}+\frac{(x-z)^j}{2(x-z)^2}$ and $P^j=\frac{(z-w)^j}{(z-w)^2}-\frac{(x-w)^j}{2(x-w)^2}$ (as in Group II), the brackets of the first term are $\left[(x-z)\cdot P-\frac{(x-z)^2-(x-w)^2}{2(z-w)^2}\right]$, $\left[(x-w)\cdot V-\frac{(x-z)^2-(x-w)^2}{2(z-w)^2}\right]$, $(x-w)^{k'}V^m$ and $(x-z)^mP^{k'}+(x-w)^mV^{k'}$. The bracket of the second and third terms is $(x-z)^{k'}P^m$.
- The third term has no $p^+$-dependent phase, so its $p^+$ integral can be done:

$$
\begin{aligned}
&\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{(x-w)^2}{(x-z)^2(p^++k^+)\left[p^+(x-z)^2+k^+(x-w)^2\right]}\\
&=\frac{1}{2\pi k^+}\,\frac{(x-w)^2}{(x-z)^2\left[(x-z)^2-(x-w)^2\right]}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right].
\end{aligned}
\tag{3.5}
$$

This was checked numerically. The third term is therefore

$$\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\frac{(w'-z)^m(x'-z)^{k'}}{(w'-z)^2(x'-z)^2}$$
$$\times\frac{(x-w)^2\,(x-z)^{k'}}{(x-z)^2\big[(x-z)^2-(x-w)^2\big]}\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]$$
$$\times\left[\frac{(z-w)^m}{(z-w)^2}-\frac{(x-w)^m}{2(x-w)^2}\right]\left[f^{c'd'a}U^{c'e'}(z)U^{d'b}(z)\rho^{e'}(x')-f^{c'd'a}U^{d'b}(z)U^{c'e'}(x')\rho^{e'}(x')\right]$$
$$\times\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{cb}(z)U^{de}(x)\rho^e(x)\right].$$

The pole at $(x-z)^2=(x-w)^2$ is removable, because the logarithm vanishes there. The kernel is integrable at $x\to z$ (since $\log\kappa$ is finite) and at $z\to w$.

- The first and second terms keep $e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}$, so their $p^+$ integral is left undone. They are finite:
  - every $\frac{(x-z)^2-(x-w)^2}{(z-w)^2}$ is $O(1/|z-w|)$;
  - at $x\to w$ the $\frac{1}{p^+(x-z)^2+k^+(x-w)^2}$ is $\frac{1}{k^+(x-w)^2}$ only at $p^+=0$, and after $\int dp^+$ it becomes $\frac{1}{(x-z)^2}\log\frac{1}{(x-w)^2}$: integrable;
  - the plus-prescribed pieces are regular at $p^+\to 0$.

  Rows V and VI are of the same type as Row III, first part. They are integrable term by term.

### 3.4 Row IV

NA

### 3.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\left[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[U^{c'e'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]\tag{3.6}$$

### 3.6 Row VI

$$
\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&+\ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.\\
&\left.-\ \frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&-\ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big] \qquad (3.7)
\end{aligned}
$$